

Ramakrishna Saravanan

Email: ramakrishna1106s@gmail.com

Phone: +1 2243824456

GitHub: <https://github.com/RickuSicku/>

LinkedIn: www.linkedin.com/in/ramakrishna-saravanan-146078230

EDUCATION

- **Northwestern University** Evanston, IL, USA
Master of Science in Artificial Intelligence Sep 2025 - Dec 2026 (Expected)
- **Indian Institute of Technology(IIT) Patna** Patna, India
B.Tech in Engineering Physics. (CPI: 8.03) Jul 2021 - May 2025

EXPERIENCE

- **Samsung Research Institute Bangalore** Jul '24 - Mar '25
Visual Intelligence Team, SRI-B
 - Developed an on-device inference model to aid in thumbnail and keyframe extraction, enabling automatic selection of aesthetically pleasing and diverse frames to summarize a movie's content best.
 - Optimized the model pipeline to achieve an state of the art inference speed of 10% of video length and performance on par with time complex deep learning solutions.
- **SPARK Summer Research Internship** May '24 - Jul '24
Under Prof. Mayank Goswami, IIT Roorkee
 - Streamlined the defect identification process by integrating advanced imaging algorithms within the Tkinter application, resulting in a significant decrease in average processing time in manual data annotation from initial analysis to final reporting for researchers handling complex metal-ceramic samples.
 - Enhanced the performance of a U-Net model in automation of defect analysis by meticulously training and fine-tuning, achieving an impressive segmentation accuracy of 97.3% for over 24,600 pixel-like defects in high-resolution CT imaging data

PROJECTS

- **PINN Simulation and Interpretability** Github Jul '24 - May '25
Under Prof. Ajay D Thakur, IIT Patna
 - Implemented Physics-Informed Neural Networks (PINNs) for solving fluid flow through porous media (Brinkman–Forchheimer equation), validating results with numerical simulations.
 - Applied saliency mapping and weight matrix pruning to enhance interpretability in deep neural networks for breast cancer detection, demonstrating how sparsity and gradient-based attention shape model decisions.
- **Ethereum Phishing Scam Detection** Github Jul '23 - May '24
Under Prof. Joydeep Chandra, IIT Patna
 - Developed and trained a Graph Convolutional Network (GCN) using Pytorch Geometric on Ethereum blockchain transaction data, achieving 98% accuracy in detecting phishing accounts.
 - Implemented preprocessing pipelines including data cleaning, feature engineering of graph node attributes, and utilizing SMOTE to perform data augmentation to solve class imbalance.

CERTIFICATIONS

- **Machine Learning Specialization , Deep Learning Specialization , Natural Language Processing**

SKILLS/FRAMEWORKS

Programming Languages: Python, SQL, R, MATLAB.

Relevant Skills / Frameworks : Tensorflow, Pytorch, PySpark, Tableau, Power BI, Scikit-learn, NumPy, Pandas, Matplotlib, Statistical Analysis, Snowflake, Data Warehousing, Data Mining, PostgreSQL, AWS, Git.